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EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. in Mechanical Engineering

Sep 2022 – June 2027 (Expected)

- **Cumulative GPA: 17.19 / 20.00** (3.62 / 4.00) | **Upper-Division GPA: 18.80 / 20.00** (3.95 / 4.00)
- **National University Entrance Exam (Konkur):** Ranked **84th** among ~150,000 candidates nationwide.
- **Selected Key Coursework:**
 - **Heat Transfer I:** 20.0 / 20.0
 - **Automatic Control:** 19.5 / 20.0
 - **Robotics:** 19.0 / 20.0
 - **Vehicle Chassis Design:** 19.4 / 20.0
 - **Internal Combustion Engines:** 19.4 / 20.0
 - **Mechanical Vibrations:** 18.9 / 20.0
- **Current Coursework (Fall 2026):** Modern Control Systems Design, Instrumentation & Control Systems, Automated Control Systems Lab.

RESEARCH INTERESTS

Core Fields: Vehicle Dynamics & Multi-Body Simulation, Automotive Control Systems, Powertrain Energy Management, Robotics & Mechatronics, Industrial Automation, Mechatronic System Integration.

SELECTED ENGINEERING PROJECTS

Modular High-Angle Drift Steering Kit & Ackermann Optimization

Feb 2026 – June 2026

Tools: CarSim, SolidWorks, Geomagic, Reverse Engineering

Project Lead

- Reverse-engineered OEM Nissan 350Z steering knuckle geometry from 3D scan point clouds using Geomagic, capturing critical suspension hardpoints without modifying chassis mounting geometry.
- Engineered a modular steering adapter with an obround-slot rotating insert, facilitating continuous geometric transition between parallel and reverse Ackermann setups.
- Redesigned and structurally reinforced the front crossmember to relocate the steering rack forward, eliminating tie-rod mechanical interference and achieving a maximum steering lock of 41° .
- Conducted multi-body vehicle dynamics simulations in CarSim (10m radius corner, constant throttle, locked differential) to analyze yaw rate, sideslip angle (β), tire slip angles, and steering torque across configurations.
- Manufactured functional prototypes via CNC machining and laser cutting/bending; verified kinematic clearances and static wheel angles ($\Delta\delta = -3^\circ$) on a test vehicle platform.

Industrial Hydraulic Power Unit (HPU) & Manifold Architecture

Spring 2026

Tools: SolidWorks, FESTO FluidSIM, Bosch Rexroth Catalogs

Academic Project

- Designed and sized an industrial HPU system driving three double-acting cylinders at 150 bar continuous pressure, ensuring cylinder stability via Euler column-buckling analysis ($P_{crit} = 249kN$).
- Matched an 11-kW powertrain integrating a Casappa Polaris gear pump, Motogen 132M4B 3-phase electric motor, and KTR Rotex flexible jaw coupling sized for 75 N·m operating torque.
- Engineered a compact vertical stack (sandwich) manifold assembly incorporating Rexroth 4/3 directional spool valves, pilot-operated checks, and meter-out throttle control valves.
- Modeled a custom reservoir in SolidWorks featuring internal thermal baffles, sediment rat holes, suction strainer, breather filter, and custom cast bellhousing with British Standard Pipe (BSP) rigid lines.

1/2-Scale Electric Drift Vehicle Powertrain & Chassis

Summer 2026 – Present

Tools: SolidWorks, Drivetrain Modeling, Longitudinal Dynamics

Personal Engineering Project

- Developing a 1/2-scale rear-wheel-drive electric vehicle architecture featuring a tubular space frame chassis, front double-wishbone suspension, and an isolated rear subframe.
- Engineered drivetrain packaging around a QS Motor 138 (7.5 kW continuous / 13.5 kW peak, 72 N·m peak torque) coupled via a 428 roller chain drive (15/42 gear ratio) directly to a solid rear axle.
- Formulated longitudinal performance models to size lithium battery pack voltage/capacity and mechanical disc brake balance for an estimated 200 kg curb weight and 90 km/h design target.

Multi-Mode HEV Energy Management Strategy Analysis

Summer 2026 – Present

Tools: MATLAB, Simulink, Stateflow (VCU Modeling)

B.Sc. Thesis Framework

- Investigated system architecture and supervisory control logic of dedicated hybrid transmission (DHT) powertrains using MATLAB/Simulink and classic Stateflow virtual vehicle environments.

- Deconstructed multi-mode vehicle control units (VCU) focusing on charge-sustaining (CS) state transitions, power-split boundaries ($P_{batt} \leq 40kW$), and engine-motor torque distribution logic.
- Formulating an undergraduate thesis framework to benchmark online rule-based supervisory strategies against deterministic optimal control formulations (Dynamic Programming / ECMS) over standard drive cycles.

PUMA Robotic Manipulator Kinematic Control

Tools: MATLAB, Simulink (Robotics Toolbox)

Spring 2025

Course Project

- Formulated forward and inverse kinematics using Denavit-Hartenberg (D-H) parameters for a 3-DOF spatial robotic arm configuration.
- Implemented trajectory generation and independent joint position controllers in MATLAB/Simulink, evaluating tracking accuracy across the operational workspace.

ADDITIONAL TECHNICAL PROJECTS

- **2D Numerical Heat Conduction Simulation (Spring 2025):** Developed a Python code to compute steady-state thermal distribution over a 2D domain with three insulated boundaries, mixed convective surface cooling, and a constant internal heat-flux source; validated finite-difference convergence (Course Grade: 20.0/20.0).
- **Progressive Blanking & Piercing Press Die (Fall 2025):** Designed complete sheet metal die tooling in SolidWorks for 2mm St12 steel using FIBRO catalog standards; optimized nesting (-50° angle) achieving 53.4% stock utilization.
- **Two-Stage Helical Reduction Gearbox (Fall 2024):** Designed an 11-kW, 11.52:1 speed reduction gearbox; performed shaft fatigue calculations, SKF bearing life selection, housing packaging, and full 2D manufacturing drafting.
- **Industrial Pneumatic Network (Spring 2026):** Sized a centralized factory compressed air ring main ($0.794m^3/min$ FAD), selecting an Atlas Copco GA11 screw compressor, refrigerant dryers, and 50 gooseneck drop stations.

INDUSTRIAL EXPERIENCE

Durali System Design and Automation (DSDA)

Engineering Intern

Tehran, Iran

Summer 2025

- Conducted CFD leakage mass-flow simulations of non-contacting labyrinth seals using ANSYS Fluent under specified operational pressure differentials.
- Evaluated structural integrity and stress concentrations of rotating shafts containing internal high-pressure fluid conduits via ANSYS Mechanical.
- Researched and modeled crowned spline geometries in SolidWorks to mitigate stress peaks induced by shaft angular eccentricity; selected materials based on yield limits and machinability.
- Assisted senior engineers in assembling, mounting, and verifying hydraulic block assemblies and directional control valves according to hydraulic schematics.

TEACHING EXPERIENCE

Sharif University of Technology

Teaching Assistant – Engineering Drawing & CAD

Tehran, Iran

Spring 2025

- Instructed 30 undergraduate students in technical projection methods, GD&T standards, and advanced part/assembly modeling workflows in SolidWorks; prepared supplementary video tutorials.

Sharif University of Technology

Teaching Assistant – Heat Transfer I

Tehran, Iran

Fall 2025

- Led weekly recitation sessions for 25 students focusing on analytical conduction/convection problem solving; managed assignment grading and coursework support.

TECHNICAL SKILLS

CAD & Mechanical Design: SolidWorks (Part, Advanced Assembly, Weldments, Sheet Metal), Reverse Engineering, Geomagic, GD&T, Machine Elements, DFM.

Vehicle & Dynamics Simulation: CarSim (Vehicle Dynamics, Handling Simulation), Multi-Body Kinematics.

Control & Mathematical Modeling: MATLAB, Simulink, Stateflow, State-Space Modern Control, Trajectory Planning.

CAE / Multiphysics: ANSYS Fluent (CFD), ANSYS Mechanical (FEA), FESTO FluidSIM.

Programming & Tools: Python (NumPy, SciPy), Git, GitHub, Microsoft Office.

Fluid Power Standards: Bosch Rexroth Catalogs, Industrial Valve Manifolds, ISO/BSP Piping, Pneumatic Ring Mains.

LANGUAGES

Persian: Native | **English:** Professional Working Proficiency | **Arabic:** Basic.